

KARACHI AQI PREDICTOR

End-to-End 72-Hour Air Quality Forecasting System

10PEARLS INTERNSHIP PROJECT REPORT

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Project: Karachi AQI Predictor
Location: Karachi, Pakistan
Final target: 0-500 AQI | **Forecast horizon:** 72 hours

Model	Recursive Random Forest
Feature schema	163 engineered features
Historical data	2+ years of hourly AQI + weather
Validation	Day 1 R ² 0.8074 Day 2 R ² 0.9677 Day 3 R ² 0.7725
Explainability	SHAP feature attribution
Deployment	Public Streamlit dashboard + Flask API

Repository: github.com/maryam-channa/karachi-aqi-predictor
Live dashboard: karachi-aqi-intelligence.streamlit.app

Final validation requirement: $R^2 > 0.70$ for each of the three 24-hour forecast blocks.

This report documents the final implementation, engineering decisions, validation results, deployment, automation, limitations, and future work for the internship deliverable.

1. Executive Summary

Karachi AQI Predictor is an end-to-end machine-learning system that forecasts Karachi's Air Quality Index for the next 72 hours. The implementation combines live operational pollution and weather inputs, a verified two-plus-year historical dataset, feature engineering, model training, recursive inference, SHAP explanations, Hopsworks integration, GitHub Actions automation, and a public Streamlit dashboard.

The final forecasting engine is a one-hour-ahead Random Forest regressor applied recursively for 72 hourly steps. Its 163-feature input schema includes current AQI and pollutant state, weather variables, lagged values, rolling statistics, cyclical time encodings, and AQI trend slopes.

The final chronological hold-out evaluation produced R^2 values of **0.807421** for Day 1, **0.967712** for Day 2, and **0.772471** for Day 3. All three daily blocks exceed the required threshold of 0.70.

The public dashboard uses packaged local final-model artifacts and current operational inputs for inference. Hopsworks is used as part of the feature-store and model-registry infrastructure. This distinction is intentional: the report does not claim that the public dashboard depends on the unavailable Hopsworks model-serving deployment.

2. Project Scope and Deliverables

Deliverable	Final implementation
Historical dataset	2+ years of hourly AQI/pollution + weather
AQI target	Project-defined 0-500 scale
ML model	Recursive Random Forest, 163 features
Forecast	72 hourly predictions aggregated into 3 daily blocks
Explainability	SHAP for first recursive forecast step
Automation	GitHub Actions: hourly feature pipeline + daily training
Data/ML platform	Hopsworks Feature Store + Model Registry
Interface	Public Streamlit dashboard + Flask API

3. Dataset and Data Acquisition

The historical pollution dataset contains 16,969 hourly observations from September 2024 through September 2026, while the historical weather backfill contains 17,520 complete hourly records covering the same period. After timestamp alignment, the combined AQI + weather dataset contains 16,968 rows.

OpenWeather supplies operational air-pollution and current-weather information. Historical weather was obtained from the Open-Meteo archive after testing showed that the required OpenWeather historical-weather access was not available for the project account. The source difference is documented rather than hidden.

Dataset	Rows	Coverage / quality
Historical pollution/AQI raw	16,969	18 temporal gaps; 552 missing hours between records
Historical weather	17,520	0 hourly gaps; 0 missing values
Combined AQI + weather	16,968	Required fields complete

3.1 Data Quality

- Duplicate timestamps: 0 in the final merged dataset.
- Required field missing values: 0 in the final combined dataset.
- Temporal gaps are preserved; feature windows are not built across discontinuities.
- Live operational observations are added to a separate local cache before optional Hopsworks persistence.

3.2 AQI Construction

The application uses a project-defined 0-500 AQI representation instead of OpenWeather's native 1-5 qualitative index. The same breakpoint and rolling-window logic used to create the historical target is applied to live pollutant concentrations when enough continuous history is available. The composite AQI is the maximum valid pollutant-specific sub-index and is clipped to 0-500.

Historical AQI summary: minimum 4, maximum 500, mean 113.2429, median 86, with 79 observations at AQI 500 and 433 observations at AQI 300 or above.

4. Pre-processing and Feature Engineering

Pre-processing is time-aware because this is a forecasting problem. Timestamps are normalized to UTC, data are sorted chronologically, duplicate hours are removed, and numerical fields are coerced to numeric values. Genuine gaps are retained and treated as boundaries for feature-window construction.

The final model uses exactly 163 features. The saved model metadata is used to verify the trained schema before inference so that the feature order in the application matches the training artifact.

Feature group	Examples	Purpose
Current state	AQI, PM2.5, PM10, NO2, O3, weather	Represent present conditions
Calendar	hour, day_of_week, month, day_of_year	Capture calendar effects
Cyclical	sin/cos encodings	Represent periodic time smoothly
Lags	1, 2, 3, 6, 12, 24 h	Capture recent temporal dependence
Rolling statistics	3, 6, 12, 24 h mean/std	Capture local level and volatility
AQI trend	6, 12, 24 h slopes	Capture recent direction of change

4.1 Inputs to the final model

AQI, PM2.5, PM10, NO2, O3, temperature, relative humidity, surface pressure, wind speed, and cloud cover are represented in the final recursive feature state. The combination allows the model to capture both persistence and environmental context.

4.2 Imputation

A median SimpleImputer is applied to the final feature matrix as part of the saved inference pipeline. The API and dashboard both use the same final model artifacts.

4.3 Preventing fabricated history

The feature builder requires genuinely available continuous hourly history. When a required window is not contiguous, it defers feature construction instead of inventing missing values. This behavior is especially important for automated hourly operation.

5. ML Model Architecture and Training

The final system uses a one-step Random Forest regression model. The regressor predicts the next hourly AQI, and the output is inserted into the next recursive state to generate the following hour. This architecture provides one coherent model across the entire 72-hour forecast.

Stage	Input	Processing	Output
Acquisition	OpenWeather + historical sources	Fetch and normalize observations	Time-stamped records
Backfill	Historical date ranges	Chunked retrieval + validation	2+ year datasets
Features	AQI/pollution + weather history	163-feature construction	Model-ready vector
Training	Features + target	Chronological hold-out + RF	One-step model
Forecast	Latest state + operational inputs	Recursive inference ×72	72 hourly predictions
Registry	Final artifact	Hopsworks registration	Versioned model
XAI	First recursive step	SHAP attribution	Feature explanation
UI	Forecast + current state	Streamlit rendering	Public dashboard

5.1 Final model configuration

Parameter	Value
Estimator	RandomForestRegressor
n_estimators	500
max_features	sqrt
min_samples_leaf	2
random_state	42
n_jobs	-1
Imputer	SimpleImputer(strategy='median')
Forecast horizon	72 hourly steps

6. Model Selection and Performance Evaluation

Several formulations were evaluated before selecting the recursive design. The decision criterion was the complete three-day requirement rather than the best result at a single horizon.

Experiment	Day 1 R ²	Day 2 R ²	Day 3 R ²	Decision
98-feature 3-day RF	0.909922	0.293603	-0.080794	Rejected: weak long horizon
364-feature weather benchmark	0.906743	0.438494	0.195949	Rejected: insufficient long-horizon performance
Direct t+24/t+48/t+72	0.614216	0.241942	0.013594	Rejected: requirement not met
Daily persistence baseline	0.526638	-0.083125	-0.438221	Baseline only
Final recursive RF	0.807421	0.967712	0.772471	Selected: all three pass

6.1 Final validation metrics

Horizon	RMSE	MAE	R ²	Status
Day 1 (0-24 h)	0.5054	0.4320	0.8074	PASS
Day 2 (24-48 h)	0.4243	0.3620	0.9677	PASS
Day 3 (48-72 h)	0.5999	0.4966	0.7725	PASS

The one-step chronological test achieved RMSE = 1.585224, MAE = 0.580212, and R² = 0.993270. These values describe the one-step driver model and are not the same as the three recursive daily metrics.

A higher Day 2 R² than Day 1 is mathematically possible because R² depends on residual error relative to the variance of the actual values within each block. The three daily scores are calculated independently.

Validation is chronological and uses a controlled historical setting in which future pollutant/weather fields are known while AQI is recursively replaced by the model prediction. This is not identical to a blind real-world 72-hour forecast.

7. Deployment, Dashboard and Explainability

The public Streamlit dashboard presents the live AQI state, weather conditions, three daily forecast summaries, the full hourly forecast, validation results, SHAP explanations, pollution profiles, historical analytics, and the final system architecture.

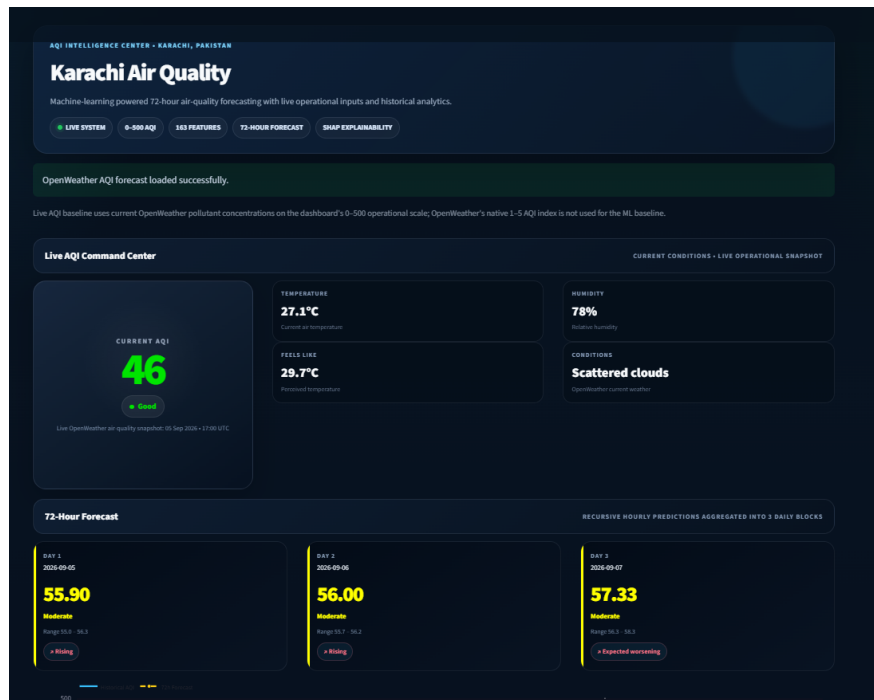


Figure 1. Latest dashboard snapshot showing the live operational AQI state and current 72-hour forecast dates.

7.1 Current deployment behavior

The current dashboard uses live OpenWeather pollutant concentrations for the operational baseline and generates the 72-hour forecast from the final local recursive model artifacts. The screenshot shows a live timestamp of 05 Sep 2026 17:00 UTC with forecast cards dated 05, 06, and 07 Sep 2026.

7.2 SHAP explainability

SHAP explains the first recursive forecast step. The dashboard reports the most influential features and their attribution direction. These values are treated as model explanations rather than causal claims.

8. Automation, Hopsworks and API

Hopsworks is integrated for feature-store/model-management infrastructure, while local artifacts provide reliable inference for the public dashboard and API. The final recursive model is registered as **karachi_aqi_recursive_random_forest_72h**, version 1, with 163 features.

Property	Verified value
Registered model	karachi_aqi_recursive_random_forest_72h
Registry	Hopsworks Model Registry
Version	1
Feature count	163
Target scale	0-500 AQI
Artifact	recursive_random_forest_compressed.pkl

8.1 GitHub Actions

Trigger	Job	Purpose
Hourly 0 * * * *	Hourly Feature Pipeline	Collect live data; update local cache; optionally persist to Hopsworks
Daily 0 0 * * *	Daily Training Pipeline	Train, validate, save artifact, register final model
Push to main	Hourly Feature Pipeline	Validate feature-pipeline changes
workflow_dispatch	Configured manually	Run pipeline on demand

The hourly pipeline was made resilient to Hopsworks compute/materialization constraints. Local cache update is performed first, while Hopsworks persistence is best-effort. This prevents a remote materialization failure from turning a successful hourly data refresh into a failed CI run.

8.2 Flask API

The repository API now loads the same final 163-feature Recursive Random Forest artifacts. Local verification returned HTTP 200 for both /health and /predict, with the response reporting 163 features and a 0-500 scale.

9. Engineering Challenges and Solutions

Challenge	Solution
Historical OpenWeather weather access unavailable for the tested subscription	Backfilled hourly weather from Open-Meteo and documented the source choice
Pollution history contained genuine temporal gaps	Preserved gaps and blocked feature windows that crossed discontinuities
Direct multi-day models degraded at longer horizons	Evaluated alternatives and selected recursive Random Forest
Hopsworks materialization/compute constraints	Made Hopsworks persistence best-effort and retained local operational cache
OpenWeather native AQI is 1-5 while final target is 0-500	Implemented the project's pollutant-based 0-500 AQI construction
Final model artifact was large	Compressed Random Forest artifact for repository/deployment practicality

10. Limitations and Methodological Caveats

- The source pollution history contains 18 temporal gaps totaling 552 missing hours.
- Historical weather and operational pollution originate from different APIs, so their statistical distributions may not be identical.
- The historical recursive validation uses known future pollutant/weather fields; real-world forecasts must also predict those inputs.
- The public dashboard uses packaged local model artifacts for inference rather than direct live Feature View inference.
- Hopsworks registration succeeded, but automatic schema/provenance inference was incomplete because a Feature View and training dataset version were not supplied.
- The 0-500 AQI is the implemented project representation and should not be described as identical to a specific government AQI methodology without separate verification.

11. Security and Reliability

- OpenWeather and Hopsworks credentials are supplied through deployment/runtime secrets rather than committed into source control.
- The public dashboard is designed to remain usable when Hopsworks persistence or serving infrastructure is unavailable.

12. Conclusion and Future Improvements

Karachi AQI Predictor is a complete internship deliverable that combines data acquisition, historical backfill, reproducible feature engineering, model training, recursive 72-hour forecasting, explainability, automation, model registration, an API layer, and a public dashboard.

The final model's chronological hold-out performance is the strongest quantitative evidence of completion: Day 1 $R^2 = 0.807421$, Day 2 $R^2 = 0.967712$, and Day 3 $R^2 = 0.772471$. Every 24-hour block exceeds the required threshold of 0.70.

The final implementation also incorporates practical reliability measures: live operational dates are used on the dashboard, Hopsworks persistence is best-effort, missing history is never fabricated, the API is aligned with the 163-feature model, and GitHub Actions performs scheduled automation.

12.1 Future Improvements

- Use dedicated 72-hour weather forecasts as inputs to live recursive inference rather than controlled historical future weather.
- Add explicit Hopsworks Feature View and training-dataset provenance to registry registration.
- Introduce statistically justified prediction intervals and uncertainty monitoring.
- Add automated data-drift and model-performance monitoring.
- Expand from a single Karachi coordinate to multiple zones or city-wide spatial forecasting.
- Evaluate LSTM/GRU or other sequence models under the same leakage-controlled validation protocol.

References

- [1] 10Pearls, project brief for the Karachi AQI Predictor internship deliverable.
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- [4] Streamlit, *Deploy your app on Community Cloud*. <https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/deploy>
- [5] Project repository: <https://github.com/maryam-channa/karachi-aqi-predictor>
- [6] Public dashboard: <https://karachi-aqi-intelligence.streamlit.app/>

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End of Report